# Big problems, good answers

## Optimization · Level 4 · Lesson 4



# Today we will

- Improve a plan in small steps, and see why it can get stuck.
- Escape a stuck plan by allowing some worse moves.
- Decide when good enough is enough, and plan a project with people.

# Remember

- Local search keeps better swaps; it can stop at a local best.
- Annealing keeps some worse swaps while hot, so it can escape.
- Agree goal, fairness, stopping rule and who decides, before you start.

# Swap to improve

1. Warm-up (5 min): How many ways can 12 pupils sit?
2. Set up (7 min): Agree the cost of a seating plan.
3. Climb (10 min): Swap two cards; keep only better swaps.
4. Escape (10 min): Allow a few worse swaps, then cool.
5. Wrap-up (8 min): Compare the costs; when is it good enough?

**Prepare the pairs in advance on a card (who works together, who is chatty) so groups score plans the same way. Use made-up names if pupils might be upset by who sits where.**

# Let’s discuss

1. Why can a plan be stuck even though no swap improves it?
2. Who should decide when our timetable is good enough?
3. What rules might our data not know about?

# Exit ticket

- Write the first two steps you would take to lead an optimization project at our school.